Signals and Systems — Reference

the big picture

Everything in this course is one of two things:

- a **signal**: a quantity that changes over time, like a voltage or a sound;
- a **system**: a machine that takes a signal in and puts a signal out.

The topics build on each other in this order:

1. **How big is a signal?** Energy and power (4–5).
2. **Building blocks.** The button δ, the switch u, and the ramp r (10).
3. **Machines.** Systems, block diagrams, difference equations (11).
4. **Describing machines.** Memory, causality, stability, linearity, time invariance (12).

The link from 1 to 4: BIBO stability asks if a *bounded* input gives a *bounded* output. That is a question about signal size.

plain-English glossary

Symbol	Say it as
$\delta[n]$, $\delta(t)$	**the button**: pressed once at time 0, then released
$u[n]$, $u(t)$	**the switch**: flipped on at time 0 and left on
$r[n]$, $r(t)$	**the ramp**: the running total of the switch, $0, 1, 2, 3, \ldots$
t	the time variable; it sweeps through every time, and is gone after you integrate
T	the half-width of a window $[-T, T]$; it is the thing that goes to ∞
t_0, n_0	"t-naught": one fixed time, such as 3
N	DT window half-width ($2N+1$ samples), or a DT period
ε	"epsilon": a tiny number that we shrink toward 0
$\int \ldots dt$	add up area as t sweeps (CT)
$\sum$	add up separate terms (DT); the DT twin of $\int$
dt	an infinitely thin slice of time. Not Δt.
B	a ceiling: the input never goes above B in size

notation: CT and DT side by side

Item	Continuous time (CT)	Discrete time (DT)
signal	$x(t)$, round brackets	$x[n]$, square brackets
time variable	t, any real number	n, integers only
period	T (seconds)	N (samples, an integer)
frequency	ω (rad/s), f (Hz)	$\hat{\omega}$ (rad/sample)
energy, power	integral $\int$	sum $\sum$
rate of change	derivative $\frac{d}{dt}$	first difference $x[n]-x[n-1]$
running total	integral $\int_{-\infty}^{t}$	running sum $\sum_{k=-\infty}^{n}$
button	$\delta(t)$, area 1	$\delta[n]$, height 1

sinusoid quantities

Symbol	Name	Unit	Link
A	amplitude	same as x	peak value
ω	angular frequency	rad/s	$\omega = 2\pi f$
f	frequency	Hz = cycles/s	$f = \frac{1}{T}$
T	period	s	$T = \frac{2\pi}{\omega}$
ϕ	phase	rad	$\phi = -\omega t_0$
t_0	time shift	s	$t_0 = -\frac{\phi}{\omega}$
$\hat{\omega}$	DT frequency	rad/sample	$\hat{\omega} = \omega T_s$
T_s, f_s	sampling period, rate	s, Hz	$f_s = \frac{1}{T_s}$

trig and complex number facts

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

$\sin 0 = 0$ and $\cos 0 = 1$. Do not swap them.

sine as a cosine	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$
negative amplitude	$-A\cos\theta = A\cos(\theta+\pi)$
square of a cosine	$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$
cosine from exponentials	$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$
sine from exponentials	$\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$
size of $a+jb$	$\sqrt{a^2+b^2}$
angle of $a+jb$	$\arctan\frac{b}{a}$ (check the quadrant)
polar form	$C = \lvert C\rvert e^{j\phi}$
size of $e^{j\theta}$	1
whole turns	$e^{j2\pi k} = 1$ for each integer k

sums, integrals, limits

Item	Result
infinite geometric sum, $\lvert r\rvert < 1$	$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$
finite geometric sum, $r \neq 1$	$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$
count of terms from $-N$ to N	$2N+1$
$\int t^n\,dt$	$\frac{t^{n+1}}{n+1}$
$\int e^{at}\,dt$	$\frac{e^{at}}{a}$
$\int_0^{\infty} e^{-at}\,dt, \quad a>0$	$\frac{1}{a}$
definite integral	$F(\text{top}) - F(\text{bottom})$
fixed number over a growing T	$\lim_{T\to\infty} \frac{c}{T} = 0$
a growing T alone	$\lim_{T\to\infty} cT = \infty \quad (c>0)$
T cancels	$\frac{cT}{2T} = \frac{c}{2}$, a constant

formula sheet

1. Signals and sampling

sampling	$x[n] = x(nT_s)$
sampling rate	$f_s = \frac{1}{T_s}$

2–3. Sinusoids

CT sinusoid	$x(t) = A\cos(\omega t + \phi)$
angular frequency	$\omega = 2\pi f$
period	$T = \frac{1}{f} = \frac{2\pi}{\omega}$
phase → time shift	$t_0 = -\frac{\phi}{\omega}$
time shift → phase	$\phi = -\omega t_0$
DT sinusoid	$x[n] = A\cos(\hat{\omega} n + \phi)$
CT → DT frequency	$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$
DT → CT frequency	$f = \frac{\hat{\omega} f_s}{2\pi}$
aliases	$\hat{\omega} + 2\pi k, \quad f + k f_s$

4–5. Energy and power

CT energy (total)	$E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt$
DT energy (total)	$E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
CT power (average)	$P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2\,dt$
DT power (average)	$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} \lvert x[n]\rvert^2$
periodic CT power	$P = \frac{1}{T}\int_{\text{one period}} \lvert x(t)\rvert^2\,dt$
periodic DT power	$P = \frac{1}{N}\sum_{\text{one period}} \lvert x[n]\rvert^2$
sinusoid power	$P = \frac{A^2}{2}$
constant c power	$P = c^2$
flat piece, level A, length D	$A^2 D$
ramp $0 \to A$, length D	$\frac{A^2 D}{3}$
line $A \to B$, length D	$\frac{D}{3}\left(A^2 + AB + B^2\right)$
DT stem of height h	h^2

6. Transformations

shift	$x(t-t_0)$: right (delay) if $t_0 > 0$
reversal	$x(-t)$: mirror about $t = 0$
scaling	$x(at)$: compress if $\lvert a\rvert > 1$
point map for $x(at+b)$	$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$
undo $y(t) = x(at+b)$	$x(t) = y\left(\frac{t-b}{a}\right)$

7. Periodic signals

CT test	$x(t+T) = x(t)$
DT test	$x[n+N] = x[n]$, N an integer
DT sinusoid period	$\frac{\hat{\omega}}{2\pi} = \frac{m}{N}$ in lowest terms
sum of two periodic signals	lcm of the two periods

8. Even and odd

even test	$x(-t) = x(t)$
odd test	$x(-t) = -x(t)$
even part	$\mathcal{E}v\{x\} = \frac{x(t) + x(-t)}{2}$
odd part	$\mathcal{O}d\{x\} = \frac{x(t) - x(-t)}{2}$

9. Complex exponentials

CT form	$x(t) = Ce^{at}, \quad a = \sigma + j\omega$
real part	$\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi), \quad \phi = \angle C$
DT form	$x[n] = C\alpha^n$

10. Button, switch, ramp

DT switch $\to$ button	$\delta[n] = u[n] - u[n-1]$
DT button $\to$ switch	$u[n] = \sum_{k=-\infty}^{n} \delta[k]$
CT switch $\to$ button	$\delta(t) = \frac{du(t)}{dt}$
CT button $\to$ switch	$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$
area of the CT button	$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$
sifting (the flash)	$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$
DT sifting	$\sum_n x[n]\,\delta[n-n_0] = x[n_0]$
CT pulse from switches	$u(t-t_1) - u(t-t_2)$: on for $t_1 \le t < t_2$
DT pulse from switches	$u[n-a] - u[n-b]$: on for $a \le n \le b-1$
any DT signal as buttons	$x[n] = \sum_k x[k]\,\delta[n-k]$
ramp	$r(t) = t\,u(t), \quad r[n] = n\,u[n]$
ramp and switch, CT	$\frac{dr}{dt} = u(t)$
ramp and switch, DT	$r[n] - r[n-1] = u[n-1]$
graph from ramps	corner: $+(\Delta\text{slope})\,r(t-t_0)$

11–12. DT systems

unit delay D	$x[n] \to x[n-1]$
first difference	$y[n] = x[n] - x[n-1]$
3-point moving average	$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$
accumulator	$y[n] = y[n-1] + x[n]$
bounded input	$\lvert x[n]\rvert \le B < \infty$
stable if	$\lvert y[n]\rvert \le$ a finite number

system property tests

Property	Ask yourself	Yes / no example
memoryless	Does y now use only x now?	yes x^2 no $x[n-1]$
causal	Does y avoid peeking at the future?	yes $x[n-1]$ no $x[n+1]$
stable	Does a capped input give a capped output?	yes x^2 no $\sum_{k\le n} x[k]$
invertible	Can I undo it and get x back?	yes $2x$ no x^2
time invariant	Does it ignore the clock? Delay in = delay out	yes $x[n-1]$ no $n\,x[n]$
linear	Scale in $\to$ scale out; add in $\to$ add out?	yes $n\,x[n]$ no $2x+3$

which tool? — given $\to$ find

Given $\to$ find	Do this	
$x(t)$, $T_s \to x[n]$	put $t = nT_s$	(1)
$A\cos(\omega t + \phi) \to f, T, t_0$	read, then divide	(2)
$f, f_s \to \hat{\omega}$	$\hat{\omega} = 2\pi f / f_s$	(3)
dies out or finite length	energy E_∞	(4)
periodic or constant	power P_∞	(4)
a graph $\to$ energy	square each piece, add	(5)
$x(t) \to x(at+b)$	map each breakpoint	(6)
$y = x(at+b) \to x$	$x(t) = y((t-b)/a)$	(6)
two graphs $\to y = A\,x(at+b)$	match two points, solve	(6)
signal $\to$ period	$2\pi/\omega$, or reduce $\hat{\omega}/2\pi$	(7)
signal $\to$ even, odd, neither	compare $x(-t)$ with $x(t)$	(8)
$Ce^{at} \to$ real signal	Euler, then the real part	(9)
integral with $\delta(t-t_0)$	the flash: three questions	(10)
a pulse graph $\to$ formula	a switch on, a switch off	(10)
DT stems $\to$ buttons	one $x[k]\,\delta[n-k]$ per stem	(10)
graph with corners $\to$ formula	a ramp per corner	(10)
blocks $\to$ one equation	substitute block by block	(11)
equation $\leftrightarrow$ block diagram	one D per shift	(11)
system $\to$ properties	ask each question	(12)
system $\to$ stable?	cap the input with B	(12)

The number in parentheses is the concept section.

Signals and Systems — Concepts

1. signals and sampling

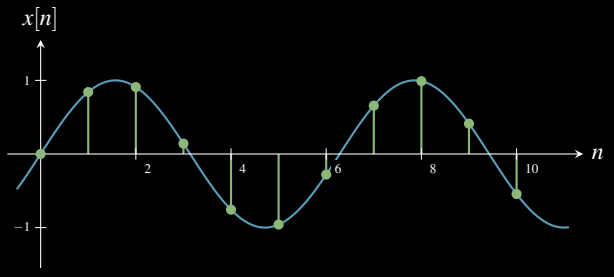
THE IDEA
A **signal** is anything that changes over time and carries information: a voltage, a sound, a temperature. **Sampling** is taking snapshots: one photo every T_s seconds, like a strobe light.

- **Continuous time** (CT): $x(t)$, round brackets. t can be any real number. The movie.
- **Discrete time** (DT): $x[n]$, square brackets. n is an integer only. The stack of photos.

Trap: $x[2.5]$ does not exist. Between two samples, a DT signal is not zero. It is simply not defined.
From movie to photos: photo number n is taken at time $t = nT_s$, so

$$x[n] = x(nT_s)$$

T_s is the **sampling period** (seconds between photos). $f_s = \frac{1}{T_s}$ is the **sampling rate** (photos per second).



The curve is the movie (CT). The stems are the photos (DT).

Worked: sample $x(t) = t^2$ with $T_s = 0.5$ s

1. PUT $t = nT_s$
$x[n] = x(0.5n) = (0.5n)^2$

↓

2. SIMPLIFY
$(0.5n)^2 = 0.5^2 \cdot n^2 = 0.25n^2$

↓

3. EVALUATE EACH SAMPLE
$x[0] = 0, \quad x[1] = 0.25, \quad x[2] = 0.25 \cdot 4 = 1, \quad x[3] = 0.25 \cdot 9 = 2.25$

↓

4. CHECK ONE SAMPLE
$n = 2$ is $t = 2 \cdot 0.5 = 1$ s. $\quad x(1) = 1^2 = 1$ ✓

Worked: (reverse) $x[n] = 0.25n^2$ came from $T_s = 0.5$ s. Find $x(t)$

1. SOLVE $t = nT_s$ FOR n
$n = \dfrac{t}{T_s} = \dfrac{t}{0.5} = 2t$

↓

2. SUBSTITUTE
$x(t) = 0.25(2t)^2$

↓

3. SIMPLIFY
$0.25 \cdot 4t^2 = t^2$

2. sinusoids

THE IDEA
Picture a wheel spinning at a steady speed, with a dot on its rim. The dot's height over time is a sinusoid. How big the wheel is: **amplitude**. How fast it spins: **frequency**. Where the dot starts: **phase**.

$$x(t) = A\cos(\omega t + \phi)$$

- A: **amplitude**. The peak. The signal swings between $-A$ and A.
- ω: **angular frequency**, in rad/s. How many radians the wheel turns each second. $\omega = 2\pi f$.
- f: **frequency**, in Hz. Full turns per second.
- $T = \frac{1}{f} = \frac{2\pi}{\omega}$: the **period**. Seconds for one full turn.
- ϕ: **phase**, in radians. A head start (or late start) for the dot.

Phase as a time shift: a late start is the same as sliding the whole wave right.

$$A\cos(\omega t + \phi) = A\cos\big(\omega(t - t_0)\big), \qquad t_0 = -\frac{\phi}{\omega}$$

A negative ϕ gives $t_0 > 0$: the wave is **delayed** (it moves right).
Trap: Put the signal in the form $A\cos(\omega t + \phi)$ before you read off the parts. Change a sine with $\sin\theta = \cos(\theta - \frac{\pi}{2})$. Change a negative A with $-A\cos\theta = A\cos(\theta + \pi)$.

Which way? — between the sinusoid numbers

Given → find	Do this
$\omega \to f$	divide by 2π: $f = \dfrac{\omega}{2\pi}$
$f \to \omega$	multiply by 2π: $\omega = 2\pi f$
$f \to T$	reciprocal: $T = \dfrac{1}{f}$
$T \to \omega$	$\omega = \dfrac{2\pi}{T}$
$\phi \to t_0$	$t_0 = -\dfrac{\phi}{\omega}$
$t_0 \to \phi$	$\phi = -\omega t_0$

Worked: read the parts of $x(t) = 3\cos\left(4\pi t - \frac{\pi}{3}\right)$

1. MATCH THE FORM $A\cos(\omega t + \phi)$
$A = 3, \quad \omega = 4\pi$ rad/s, $\quad \phi = -\frac{\pi}{3}$ rad

↓

2. FREQUENCY
$f = \dfrac{\omega}{2\pi} = \dfrac{4\pi}{2\pi} = 2$ Hz

↓

3. PERIOD
$T = \dfrac{1}{f} = \dfrac{1}{2} = 0.5$ s

↓

4. TIME SHIFT
$t_0 = -\dfrac{\phi}{\omega} = -\dfrac{-\pi/3}{4\pi} = \dfrac{\pi/3}{4\pi} = \dfrac{1}{3 \cdot 4} = \frac{1}{12}$ s

↓

5. REWRITE AS A SHIFT
$x(t) = 3\cos\left(4\pi(t - \frac{1}{12})\right)$: delayed by $\frac{1}{12}$ s

↓

6. CHECK
$4\pi \cdot \frac{1}{12} = \frac{\pi}{3}$. So $4\pi(t - \frac{1}{12}) = 4\pi t - \frac{\pi}{3}$ ✓

Worked: (sine first) read the parts of $x(t) = 4\sin(10t)$

1. CHANGE SINE TO COSINE
$\sin\theta = \cos(\theta - \frac{\pi}{2})$, so $x(t) = 4\cos\left(10t - \frac{\pi}{2}\right)$

↓

2. MATCH THE FORM
$A = 4, \quad \omega = 10$ rad/s, $\quad \phi = -\frac{\pi}{2}$

↓

3. FREQUENCY AND PERIOD
$f = \dfrac{10}{2\pi} = 1.59$ Hz. $\quad T = \dfrac{2\pi}{10} = 0.628$ s

Worked: (reverse) build $x(t)$ from $A = 2$, $T = 0.4$ s, first peak at $t = 0.1$ s

1. PERIOD TO ω
$\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{0.4} = 5\pi$ rad/s

↓

2. PEAK TIME IS THE SHIFT
A cosine peaks where its angle is 0. The first peak is at $t = 0.1$, so $t_0 = 0.1$ s.

↓

3. SHIFT TO PHASE
$\phi = -\omega t_0 = -5\pi \cdot 0.1 = -0.5\pi = -\frac{\pi}{2}$

↓

4. ASSEMBLE
$x(t) = 2\cos\left(5\pi t - \frac{\pi}{2}\right)$

↓

5. CHECK AT $t = 0.1$
Angle: $5\pi(0.1) - \frac{\pi}{2} = \frac{\pi}{2} - \frac{\pi}{2} = 0$. $\quad 2\cos 0 = 2$, the peak ✓

3. DT sinusoids and DT frequency

THE IDEA
Photograph the spinning wheel with a strobe. In the photos, all you can see is *how far the wheel turned between photos*. That angle per photo is the **DT frequency** $\hat{\omega}$, in radians per sample.

Sample $A\cos(\omega t + \phi)$ at $t = nT_s$:

$$x[n] = A\cos(\omega n T_s + \phi) = A\cos(\hat{\omega} n + \phi), \qquad \hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$$

Aliasing is the wagon-wheel effect in old films. A wheel that turns $\frac{\pi}{4}$ between photos looks identical to one that turns $\frac{\pi}{4} + 2\pi$: the extra full turn is invisible. So $\hat{\omega}$ and $\hat{\omega} + 2\pi$ give the same photos, and f and $f + f_s$ give the same samples.

Which way? — CT frequency or DT frequency

Given → find	Do this
$f, f_s \to \hat{\omega}$	$\hat{\omega} = \dfrac{2\pi f}{f_s}$
$\omega, T_s \to \hat{\omega}$	$\hat{\omega} = \omega T_s$
$\hat{\omega}, f_s \to f$	$f = \dfrac{\hat{\omega} f_s}{2\pi}$ (the lowest answer)

Trap: DT to CT has more than one answer. Any alias $f + k f_s$ fits the same photos.

Worked: sample $x(t) = \cos(200\pi t)$ at $f_s = 800$ Hz

1. READ ω AND f
$\omega = 200\pi$ rad/s. $\quad f = \dfrac{200\pi}{2\pi} = 100$ Hz

↓

2. SAMPLING PERIOD
$T_s = \dfrac{1}{800}$ s

↓

3. DT FREQUENCY
$\hat{\omega} = \omega T_s = \dfrac{200\pi}{800} = \frac{\pi}{4}$ rad/sample

↓

4. THE DT SIGNAL
$x[n] = \cos\left(\frac{\pi}{4} n\right)$

Worked: (reverse) $\hat{\omega} = \frac{\pi}{4}$ at $f_s = 800$ Hz. Find f

1. FORMULA
$f = \dfrac{\hat{\omega} f_s}{2\pi}$

↓

2. SUBSTITUTE
$f = \dfrac{(\pi/4) \cdot 800}{2\pi} = \dfrac{200\pi}{2\pi} = 100$ Hz

Worked: the alias: does $f = 900$ Hz give the same samples?

1. DT FREQUENCY
$\hat{\omega} = \dfrac{2\pi \cdot 900}{800} = \dfrac{1800\pi}{800} = \dfrac{9\pi}{4}$

↓

2. REMOVE A WHOLE TURN
$\dfrac{9\pi}{4} - 2\pi = \dfrac{9\pi}{4} - \dfrac{8\pi}{4} = \dfrac{\pi}{4}$

↓

3. COMPARE
Same $\hat{\omega}$ as 100 Hz ⇒ same samples. $\quad 900 = 100 + 800 = f + f_s$.

4. energy and power

THE IDEA
Think of $|x|^2$ as **money spent per day**, forever.
- **Energy** is your total spending, over all time.
- **Power** is your average spending per day, over all time.

Spend \$100 once, then stop: the total is finite, and the average over infinitely many days is 0. Spend \$5 every day forever: the total is infinite, and the average is \$5.

Why square?

Picture x as a voltage across a $1\,\Omega$ resistor. Electrical power is $P = V \cdot I$, and Ohm's law gives $I = V/R$, so $P = V^2/R = x^2$. The signal shows up *twice*: once as the voltage, once as the current it drives.

Squaring also makes sense of the sign. A toaster heats up whichever way the current flows. A negative voltage delivers energy just like a positive one. Squaring turns every value non-negative, so negatives stop cancelling positives:

- positive stays positive: $(0.5)^2 = 0.25$
- negative turns positive: $(-0.5)^2 = 0.25$
- zero stays zero

Without squaring, $\cos(t)$ would average to 0, as if a wall outlet held nothing.

The two formulas, in English

Total energy: add up $|x|^2$ over all time. No dividing.

$$E_\infty = \int_{-\infty}^{\infty} |x(t)|^2\,dt \qquad E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average power: look through a **window** from $-T$ to T. Add up $|x|^2$ inside it, divide by the window's length $2T$, then let the window grow forever.

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2\,dt$$

In DT, the window $n = -N \ldots N$ holds $2N+1$ samples (count $-2, -1, 0, 1, 2$: that is $5 = 2(2)+1$):

$$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

Periodic shortcut: one period tells the whole story.
$P = \frac{1}{T} \int_{\text{one period}} |x(t)|^2\,dt$.

Energy vs power, side by side

	Energy	Power
means	the total	the average
divide by $2T$?	no	yes
money	lifetime spending	spending per day
constant $x = 4$	$lim 32T = \infty$	$lim \frac{32T}{2T} = 16$

Rule: a final E_∞ or P_∞ never contains T or N. If a T is still there, you have not taken the limit yet.

Which one to compute? Classify by shape first

The signal	Result
dies out, or has finite length	**energy signal**: E_∞ finite, $P_\infty = 0$
periodic, or constant	**power signal**: P_∞ finite, $E_\infty = \infty$
grows without limit ($x = t$)	**neither**: $E_\infty = \infty$, $P_\infty = \infty$

Why an energy signal has $P_\infty = 0$: its total is a fixed number E. Divide it by a window growing to ∞: $lim \frac{E}{2T} = 0$. \$50 spread over a million days is almost nothing per day.

Why a constant has $E_\infty = \infty$ but finite power: the energy in the window grows *at the same rate* as the window, so T cancels in the ratio. "c^2 per second, forever" is a *rate*, which is the power. The total is ∞.

Why you can shrink the limits: where $x = 0$, it adds nothing. Once the window covers every non-zero part, widening it only adds zeros. In DT, that happens once N reaches the stem farthest from $n = 0$, on either side.

Traps from practice

Trap: Square first. Adding raw stem values lets negatives cancel. Check: $E_\infty = 0$ is only possible if the signal is 0 everywhere.

Trap: Energy is not power. "16 forever" is the power. The energy is the total, ∞.

Trap: T is not t. t disappears when you integrate. T is what goes to ∞: write $lim_{T\to\infty}$.

Trap: $\frac{c}{T}$ is not cT. Dividing by T puts T in the denominator, and then $\frac{c}{T} \to 0$, not c.

Trap: dt, not Δt. dt also tells you which letter is the variable.

Trap: Write the label. Energy signal, power signal, or neither, not "constant" or "growth."

Worked: pulse: $x(t) = 2$ for $0 \le t \le 3$, 0 elsewhere

1. SQUARE
$|x(t)|^2 = 2^2 = 4$ on $[0,3]$. 0 elsewhere.

↓

2. LIMITS SHRINK TO WHERE $x \neq 0$
$E_\infty = \int_0^3 4\,dt$ *(outside $[0,3]$, $x = 0$ adds nothing)*

↓

3. ANTIDERIVATIVE
$F(t) = 4t$

↓

4. TOP MINUS BOTTOM
$4(3) - 4(0) = 12$ *(check: $A^2 D = 4 \cdot 3$)*

↓

5. POWER
$P_\infty = \lim_{T\to\infty} \frac{12}{2T} = 0 \Rightarrow$ energy signal

Worked: decaying exponential: $x(t) = 3e^{-2t}u(t)$

1. WHERE IS IT NOT ZERO?
The switch $u(t)$ is off for $t < 0$. So integrate from 0 to ∞.

↓

2. SQUARE
$\left(3e^{-2t}\right)^2 = 3^2 \cdot \left(e^{-2t}\right)^2 = 9e^{-4t}$ *(double the exponent)*

↓

3. SET UP
$E_\infty = \lim_{b\to\infty} \int_0^b 9e^{-4t}\,dt$

↓

4. ANTIDERIVATIVE
$F(t) = \frac{9}{-4}e^{-4t} = -2.25e^{-4t}$

↓

5. TOP ($t = b$), THEN $b \to \infty$
$F(b) = -2.25e^{-4b} \to -2.25(0) = 0$

↓

6. BOTTOM ($t = 0$)
$F(0) = -2.25e^0 = -2.25(1) = -2.25$

↓

7. TOP MINUS BOTTOM
$E_\infty = 0 - (-2.25) = 2.25 \Rightarrow$ energy signal

Worked: DT geometric: $x[n] = \left(\frac{1}{2}\right)^n u[n]$

1. WHERE IS IT NOT ZERO?
The switch $u[n]$ is off for $n < 0$. So sum from $n = 0$.

↓

2. SQUARE
$\left(\left(\frac{1}{2}\right)^n\right)^2 = \left(\frac{1}{2}\right)^{2n} = \left(\frac{1}{4}\right)^n$

↓

3. SET UP
$E_\infty = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$ *(a geometric sum with $r = \frac{1}{4}$)*

↓

4. GEOMETRIC SUM FORMULA
$\frac{1}{1-r} = \frac{1}{1-\frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$

↓

5. CHECK WITH THE FIRST TERMS
$1 + 0.25 + 0.0625 + 0.0156 = 1.328$, close to 1.333 ✓

Worked: sinusoid: $x(t) = 4\cos(3t)$

1. SQUARE
$|x|^2 = 16\cos^2(3t)$ *(never negative now)*

↓

2. USE $\cos^2\theta = \frac{1+\cos 2\theta}{2}$
$16 \cdot \frac{1+\cos(6t)}{2} = 8 + 8\cos(6t)$

↓

3. AVERAGE EACH TERM
The constant 8 averages to 8. $8\cos(6t)$ averages to 0 *(equal area above and below 0)*.

↓

4. POWER
$P_\infty = 8 + 0 = 8$

↓

5. CHECK WITH THE SHORTCUT
$\frac{A^2}{2} = \frac{4^2}{2} = \frac{16}{2} = 8$ ✓ $E_\infty = \infty \Rightarrow$ power signal

Worked: constant: $x(t) = 3$ for all t

1. SQUARE
$|x|^2 = 9$

↓

2. ENERGY IN THE WINDOW $[-T, T]$
$\int_{-T}^{T} 9\,dt = 9T - 9(-T) = 18T$ *(watch the minus-a-negative)*

↓

3. ENERGY: LET $T \to \infty$, NO DIVIDING
$E_\infty = \lim_{T\to\infty} 18T = \infty$

↓

4. POWER: DIVIDE BY $2T$, THEN THE LIMIT
$\frac{18T}{2T} = 9$, so $P_\infty = \lim_{T\to\infty} 9 = 9$ *(T cancelled)*

↓

5. LABEL
power signal *(matches $P = c^2$)*

Worked: DT power: $x[n] = (-1)^n$

1. SQUARE
$\left((-1)^n\right)^2 = 1$ for each n

↓

2. SUM FROM $-N$ TO N
$2N+1$ terms, each 1: sum $= 2N+1$

↓

3. DIVIDE AND LET $N \to \infty$
$P_\infty = \frac{2N+1}{2N+1} = 1$

5. energy from a graph

THE IDEA
Do not set up one big integral. Chop the graph into pieces, square each piece, find each piece's energy, and add. In DT there are no widths: each stem is one sample and counts once, so its energy is just its height squared.

1. **Read the graph into pieces.** Write $x(t)$ on each time range: a constant, or a straight line.
2. **Square each piece.**
3. **CT**: integrate each squared piece over its range. **DT**: square each stem height.
4. **Add the pieces.**

Straight line through (t_1, x_1) with slope $m = \frac{\Delta x}{\Delta t}$: $x(t) = x_1 + m(t - t_1)$.

Shortcuts for one piece of length D:

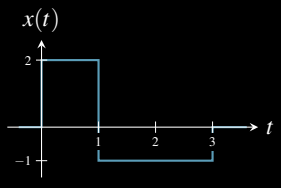
Piece of the graph	Energy of that piece
flat at level A	$A^2 D$ (a rectangle of height A^2)
ramp from 0 to A (or A to 0)	$\frac{A^2 D}{3}$
line from A to B	$\frac{D}{3}\left(A^2 + AB + B^2\right)$
DT stem of height h	h^2

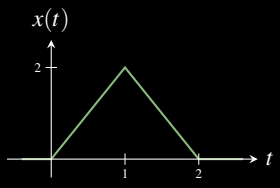
Use a shortcut to check an integral, and an integral to check a shortcut.

Trap: Negative parts count positive. $(-2)^2 = 4$. They do not cancel positive parts.

Trap: A squared ramp is a curve, not a triangle. Use $\frac{A^2 D}{3}$, not $\frac{A^2 D}{2}$.

Trap: If the graph repeats forever, $E_\infty = \infty$. Find the power over one period.





Worked: two flat pieces (left graph)

1. READ THE PIECES
$x(t) = 2$ on $[0,1]$. $x(t) = -1$ on $[1,3]$. 0 elsewhere.

↓

2. SQUARE EACH PIECE
$2^2 = 4$ on $[0,1]$. $(-1)^2 = 1$ on $[1,3]$.

↓

3. ENERGY OF PIECE 1
$\int_0^1 4\,dt = 4 \cdot (1-0) = 4$

↓

4. ENERGY OF PIECE 2
$\int_1^3 1\,dt = 1 \cdot (3-1) = 2$

↓

5. ADD
$E_\infty = 4 + 2 = 6$

Worked: a triangle (right graph)

1. RISE: FIND THE LINE
From $(0,0)$ to $(1,2)$: slope $m = \frac{2-0}{1-0} = 2$. $x(t) = 0 + 2(t-0) = 2t$ on $[0,1]$.

↓

2. RISE: SQUARE
$(2t)^2 = 4t^2$

↓

3. RISE: INTEGRATE
$F(t) = \frac{4t^3}{3}$. $F(1) - F(0) = \frac{4}{3} - 0 = \frac{4}{3}$

↓

4. FALL: FIND THE LINE
From $(1,2)$ to $(2,0)$: slope $m = \frac{0-2}{2-1} = -2$. $x(t) = 2 - 2(t-1) = 4 - 2t$ on $[1,2]$.

↓

5. FALL: SQUARE
$(4-2t)^2 = 16 - 16t + 4t^2$ *($(a-b)^2 = a^2 - 2ab + b^2$)*

↓

6. FALL: ANTIDERIVATIVE
$G(t) = 16t - 8t^2 + \frac{4t^3}{3}$

↓

7. FALL: TOP ($t = 2$)
$G(2) = 32 - 32 + \frac{32}{3} = \frac{32}{3}$

↓

8. FALL: BOTTOM ($t = 1$)
$G(1) = 16 - 8 + \frac{4}{3} = 8 + \frac{4}{3} = \frac{28}{3}$

↓

9. FALL: TOP MINUS BOTTOM
$\frac{32}{3} - \frac{28}{3} = \frac{4}{3}$

↓

10. ADD
$E_\infty = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$

↓

11. CHECK WITH THE SHORTCUT
Each ramp: $\frac{A^2 D}{3} = \frac{2^2 \cdot 1}{3} = \frac{4}{3}$ ✓

Worked: a line that does not start at 0: from 1 to 3 over $[0,2]$

1. FIND THE LINE
Slope $m = \frac{3-1}{2-0} = 1$. $x(t) = 1 + 1 \cdot (t-0) = 1+t$

↓

2. SQUARE
$(1+t)^2 = 1 + 2t + t^2$

↓

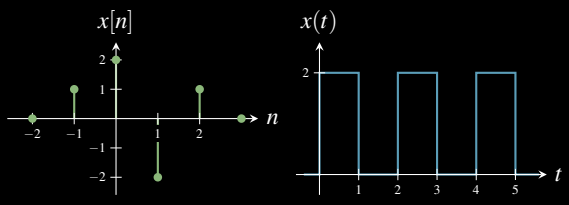
3. ANTIDERIVATIVE
$F(t) = t + t^2 + \frac{t^3}{3}$

↓

4. TOP MINUS BOTTOM
$F(2) = 2 + 4 + \frac{8}{3} = \frac{26}{3}$. $F(0) = 0$. $E = \frac{26}{3}$

↓

5. CHECK WITH THE SHORTCUT
$\frac{D}{3}\left(A^2 + AB + B^2\right) = \frac{2}{3}(1+3+9) = \frac{26}{3}$ ✓



Worked: a DT stem plot (left graph)

1. READ THE STEMS
$x[-1] = 1$, $x[0] = 2$, $x[1] = -2$, $x[2] = 1$. All others 0.

↓

2. SQUARE EACH STEM
$1^2 = 1$, $2^2 = 4$, $(-2)^2 = 4$, $1^2 = 1$ *(write every square separately)*

↓

3. ADD
$E_\infty = 1 + 4 + 4 + 1 = 10$ *(no integral: DT energy is a sum)*

↓

4. POWER
$P_\infty = \lim_{N\to\infty} \frac{10}{2N+1} = 0 \Rightarrow$ energy signal

Worked: a periodic graph (right graph): power, not energy

1. ONE PERIOD
The pattern repeats every $T = 2$ s: $x = 2$ on $[0,1]$, $x = 0$ on $[1,2]$.

↓

2. ENERGY OF ONE PERIOD
$2^2 \cdot 1 + 0^2 \cdot 1 = 4$

↓

3. DIVIDE BY THE PERIOD
$P = \frac{4}{2} = 2$

↓

4. TOTAL ENERGY
It repeats forever: $E_\infty = \infty \Rightarrow$ power signal

6. transformations of the time variable

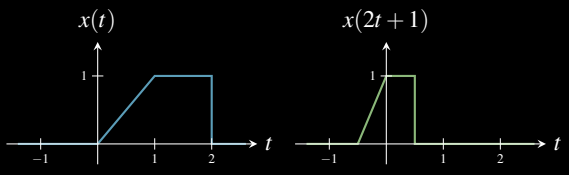
THE IDEA
You are not changing the shape, only *when* things happen. Ask one question of every corner of the graph: **where did this point move to?** It keeps its height and just lands at a new time.

- **Shift**: $x(t - t_0)$. If $t_0 > 0$, the signal moves right: a **delay**, it happens later. If $t_0 < 0$, it moves left: an **advance**, it happens earlier.
- **Reversal**: $x(-t)$. Play the tape backwards: mirror about $t = 0$.
- **Scaling**: $x(at)$. Fast-forward if $|a| > 1$ (compressed). Slow motion if $|a| < 1$ (stretched).

Point-map method for $x(at + b)$: the old time t_{old} shows up where $at_{\text{new}} + b = t_{\text{old}}$. Solve for the new time:

$$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$$

One step, no order of operations to remember.
Two-step method: first shift by b to make $x(t + b)$. Then replace t with at.
Trap: Scale first, then shift by b, is wrong. $x(2t)$ shifted left by 1 is $x(2(t+1)) = x(2t+2)$, not $x(2t+1)$.
Trap: The sign inside is opposite to the direction. $x(t-3)$ moves *right*.
Trap: DT scaling loses photos: $x[2n]$ keeps only the even-index samples.



Worked: sketch $x(2t+1)$ from $x(t)$ (left graph)

1. LIST THE BREAKPOINTS OF x
$t = 0$ (value 0), $t = 1$ (value 1), $t = 2$ (value 1, then drops to 0)

↓

2. READ a AND b
$x(2t+1)$: $a = 2$, $b = 1$

↓

3. MAP $t = 0$
$\frac{0-1}{2} = -\frac{1}{2}$

↓

4. MAP $t = 1$
$\frac{1-1}{2} = 0$

↓

5. MAP $t = 2$
$\frac{2-1}{2} = \frac{1}{2}$

↓

6. DRAW WITH THE SAME VALUES
Ramp $0 \to 1$ on $[-\frac{1}{2}, 0]$. Flat at 1 on $[0, \frac{1}{2}]$. 0 elsewhere (right graph).

↓

7. CHECK ONE POINT
At $t = 0$: $x(2 \cdot 0 + 1) = x(1) = 1$ ✓

Worked: (reversal) sketch $x(-t+2)$ from the same $x(t)$

1. READ a AND b
$a=-1,\quad b=2.\quad t_{\text{new}}=\dfrac{t_{\text{old}}-2}{-1}=2-t_{\text{old}}$

↓

2. MAP EACH BREAKPOINT
$0\to 2,\quad 1\to 1,\quad 2\to 0$

↓

3. DRAW WITH THE SAME VALUES
Flat at 1 on $[0,1]$. Ramp $1\to 0$ on $[1,2]$. The picture is mirrored.

↓

4. CHECK ONE POINT
At $t=1.5$: $x(-1.5+2)=x(0.5)=0.5$. The ramp from 1 to 0 is at 0.5 there ✓

Worked: DT: $x[0]=1$, $x[1]=2$, $x[2]=3$, else 0

1. DELAY $x[n-2]$
Each sample moves right 2: $n=2$: 1, $n=3$: 2, $n=4$: 3

↓

2. REVERSE $x[-n]$
Each sample mirrors: $n=0$: 1, $n=-1$: 2, $n=-2$: 3

↓

3. COMPRESS $x[2n]$
$n=0$: $x[0]=1$. $n=1$: $x[2]=3$. $x[1]=2$ is lost.

Which way? — undo a transformation

Given → find	Do this
$x(t)\to x(at+b)$	map each breakpoint: $t_{\text{new}}=\frac{t_{\text{old}}-b}{a}$
$y(t)=x(at+b)\to x(t)$	$x(t)=y\left(\frac{t-b}{a}\right)$

Trap: $x[2n]$ cannot be undone. The odd-index photos were thrown away.

Worked: (reverse) $y(t)=x(2t+1)$ is given. Find $x(t)$

1. NAME THE INSIDE
Let $\tau=2t+1$. Then $y(t)=x(\tau)$.

↓

2. SOLVE FOR t
$2t=\tau-1\Rightarrow t=\dfrac{\tau-1}{2}$

↓

3. SUBSTITUTE
$x(\tau)=y\left(\dfrac{\tau-1}{2}\right)$. Rename τ as t: $x(t)=y\left(\frac{t-1}{2}\right)$

↓

4. CHECK
$y\left(\frac{t-1}{2}\right)=x\left(2\cdot\frac{t-1}{2}+1\right)=x(t-1+1)=x(t)$ ✓

Which way? — find the transformation from x to y

Some problems give two graphs and ask for y in terms of x:
$y(t)=A\,x(at+b)$.

1. **Amplitude**: compare the heights. $A=\frac{\text{height of } y}{\text{height of } x}$. Upside down means a negative A.
2. **Orientation**: same way round, or mirrored? Mirrored means $a<0$.
3. **Two matching points**: pick two breakpoints of x (t_1, t_2) and the points of y they moved to (τ_1, τ_2).
4. **Solve** $a\tau_1+b=t_1$ and $a\tau_2+b=t_2$ for a and b. *(y at τ is x at $a\tau+b$)*
5. **Check** a third point.

Worked: $x(t)$ is a ramp from 0 to 1 on $[0,4]$. $y(t)$ is the same ramp, height 3, on $[-1,1]$

1. AMPLITUDE
$A=\dfrac{3}{1}=3$

↓

2. ORIENTATION
Both rise from left to right ⇒ not mirrored, $a>0$

↓

3. MATCHING POINTS
The start $t_1=0$ moves to $\tau_1=-1$. The end $t_2=4$ moves to $\tau_2=1$.

↓

4. TWO EQUATIONS
$a(-1)+b=0\quad$ and $\quad a(1)+b=4$

↓

5. SUBTRACT THE FIRST FROM THE SECOND
$a-(-a)=4-0\Rightarrow 2a=4\Rightarrow a=2$

↓

6. BACK-SUBSTITUTE
$-2+b=0\Rightarrow b=2$

↓

7. RESULT AND CHECK
$y(t)=3\,x(2t+2)$. Middle: $t=0$ gives $x(2)=\frac{1}{2}$, so $y=1.5$, half of 3 ✓

Worked: (mirrored) the same x, but y falls from 3 to 0 on $[-1,1]$

1. MATCHING POINTS
The start of the ramp ($t_1=0$, value 0) is now on the right: $\tau_1=1$. The end ($t_2=4$) is on the left: $\tau_2=-1$.

↓

2. TWO EQUATIONS
$a(1)+b=0\quad$ and $\quad a(-1)+b=4$

↓

3. SOLVE
Subtract: $2a=-4\Rightarrow a=-2$. Then $b=2$.

↓

4. RESULT
$y(t)=3\,x(-2t+2)$ *($a<0$ because the shape is mirrored)*

Worked: (DT) $x[0]=1$, $x[1]=2$, $x[2]=3$. $y[-1]=3$, $y[0]=2$, $y[1]=1$. Find y in terms of x

1. ORIENTATION
x rises to the right. y falls to the right ⇒ mirrored: $y[n]=x[-n+b]$

↓

2. ONE MATCHING POINT
$y[-1]=3=x[2]$. So $-(-1)+b=2\Rightarrow 1+b=2\Rightarrow b=1$

↓

3. RESULT
$y[n]=x[-n+1]$

↓

4. CHECK THE OTHERS
$n=0$: $x[1]=2$ ✓ $\quad n=1$: $x[0]=1$ ✓

7. periodic signals

THE IDEA
A periodic signal is a song on repeat. Slide it forward by one period and you cannot tell anything happened. The **fundamental period** is the length of the shortest loop.

$$x(t)=x(t+T)\qquad x[n]=x[n+N]$$

N must be an integer: you cannot slide by half a photo.

- **CT sinusoid** $\cos(\omega t)$: always periodic. $T=\frac{2\pi}{\omega}$.
- **DT sinusoid** $\cos(\hat{\omega}n)$: periodic only if, after some whole number of photos, the wheel has made a whole number of turns. That means $\frac{\hat{\omega}}{2\pi}$ is a fraction $\frac{m}{N}$. Reduce it to lowest terms. The denominator is N.
- **Two periodic signals added**: both loops line up again after the least common multiple (lcm) of the two periods.

Trap: $\cos(n)$ is not periodic. $\frac{1}{2\pi}$ is not a fraction of integers, so the photos never exactly repeat. The CT signal $\cos(t)$ is periodic. Sampling can destroy

periodicity.

Worked: period of $\cos\left(\frac{2\pi}{3}t\right)+\sin\left(\frac{\pi}{2}t\right)$

1. FIRST PERIOD
$T_1=\dfrac{2\pi}{2\pi/3}=2\pi\cdot\dfrac{3}{2\pi}=3$ s

↓

2. SECOND PERIOD
$T_2=\dfrac{2\pi}{\pi/2}=2\pi\cdot\dfrac{2}{\pi}=4$ s

↓

3. LEAST COMMON MULTIPLE
Multiples of 3: $3,6,9,12$. Multiples of 4: $4,8,12$. $T=12$ s

↓

4. CHECK
12 s holds $\frac{12}{3}=4$ cycles of the first and $\frac{12}{4}=3$ of the second ✓

Worked: DT period of $x[n]=\cos\left(\frac{3\pi}{8}n\right)$

1. DIVIDE BY 2π
$\dfrac{\hat{\omega}}{2\pi}=\dfrac{3\pi/8}{2\pi}=\dfrac{3\pi}{8}\cdot\dfrac{1}{2\pi}=\dfrac{3}{16}$

↓

2. LOWEST TERMS?
3 and 16 share no factor. $N=16$

↓

3. CHECK
$\dfrac{3\pi}{8}\cdot 16=6\pi=3\cdot 2\pi$, whole turns ✓

Worked: reduce first: $x[n]=\cos\left(\frac{4\pi}{6}n\right)$

1. DIVIDE BY 2π
$\dfrac{4\pi/6}{2\pi}=\dfrac{4}{12}$

↓

2. REDUCE
$\dfrac{4}{12}=\dfrac{1}{3}\Rightarrow N=3$ not 12

Worked: is $x[n]=\cos(n)$ ***periodic?***

1. DIVIDE BY 2π
$\dfrac{1}{2\pi}$

↓

2. IS IT A FRACTION OF INTEGERS?
No: π is irrational. ⇒ not periodic

8. even and odd signals

THE IDEA
Even: put a mirror on the vertical axis. The left half is the reflection of the right half. **Odd**: spin the graph half a turn about the origin, and it lands on itself. Left half is the right half, flipped *and* upside down.

- **Even**: $x(-t)=x(t)$. Examples: cos, t^2.
- **Odd**: $x(-t)=-x(t)$. Examples: sin, t^3.

Test: replace t with $-t$, simplify, and compare with $x(t)$.
Trap: An odd signal is always 0 at the origin: $x(0)=-x(0)$ forces $x(0)=0$.
Every signal splits into an even part plus an odd part:

$$\mathcal{E}v\{x\}=\frac{x(t)+x(-t)}{2}\qquad \mathcal{O}d\{x\}=\frac{x(t)-x(-t)}{2}$$

Products work like signs: even · even = even, odd · odd = even, even · odd = odd.

Worked: is $x(t)=t^3-2t$ ***even, odd, or neither?***

1. REPLACE t WITH $-t$
$x(-t)=(-t)^3-2(-t)$

↓

2. SIMPLIFY
$(-t)^3=-t^3$, $-2(-t)=2t$. $x(-t)=-t^3+2t$

↓

3. COMPARE
$-t^3+2t=-(t^3-2t)=-x(t)\Rightarrow$ odd

Worked: $x(t)=t^2+t$***: test, then split***

1. REPLACE t WITH $-t$
$x(-t)=(-t)^2+(-t)=t^2-t$

↓

2. COMPARE
$t^2-t\neq x(t)$ and $\neq -x(t)\Rightarrow$ neither

↓

3. EVEN PART
$\dfrac{(t^2+t)+(t^2-t)}{2}=\dfrac{2t^2}{2}=t^2$

↓

4. ODD PART
$\dfrac{(t^2+t)-(t^2-t)}{2}=\dfrac{2t}{2}=t$

↓

5. CHECK
$t^2+t=x(t)$ ✓

Worked: DT split: $x[0]=1$, $x[1]=2$, $x[2]=3$, ***else*** 0

1. TABLE OF $x[n]$ AND $x[-n]$, THEN THE TWO PARTS

n	-2	-1	0	1	2
$x[n]$	0	0	1	2	3
$x[-n]$	3	2	1	0	0
$\frac{x[n]+x[-n]}{2}$	1.5	1	1	1	1.5
$\frac{x[n]-x[-n]}{2}$	−1.5	−1	0	1	1.5

↓

2. HOW TO FILL $x[-n]$
Read x at the opposite index. $n=-2$: $x[2]=3$. $n=1$: $x[-1]=0$.

↓

3. CHECK EACH COLUMN
even + odd $=x[n]$. $n=2$: $1.5+1.5=3$ ✓ $n=-2$: $1.5-1.5=0$ ✓

9. complex exponentials

THE IDEA
$e^{j\theta}$ is an arrow of length 1, pointing at angle θ. Let $\theta=\omega t$ and the arrow spins. Its shadow on the horizontal axis is a cosine. Multiply by $e^{\sigma t}$ and the arrow also grows ($\sigma>0$) or shrinks ($\sigma<0$) as it spins: a spiral.

Euler's formula: $e^{j\theta}=\cos\theta+j\sin\theta$. Engineers write $j=\sqrt{-1}$.
Back to cosine and sine: $\cos\theta=\dfrac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta=\dfrac{e^{j\theta}-e^{-j\theta}}{2j}$. Two arrows spinning opposite ways: their imaginary parts cancel.
CT form: $x(t)=Ce^{at}$, with $C=|C|e^{j\phi}$ (starting length and angle) and $a=\sigma+j\omega$ (growth rate and spin rate).

Case	Real part looks like
a real ($\omega=0$)	no spin: a **real exponential**. Grows if $\sigma>0$, decays if $\sigma<0$
$a=j\omega$ ($\sigma=0$)	pure spin: a **sinusoid** $\lvert C\rvert\cos(\omega t+\phi)$, $T=\frac{2\pi}{\lvert\omega\rvert}$
general a	spiral: a **damped sinusoid** $\lvert C\rvert e^{\sigma t}\cos(\omega t+\phi)$

The $|C|e^{\sigma t}$ part is the **envelope**: the curve the peaks follow.
DT form: $x[n]=C\alpha^n$. Each step multiplies by α. $|\alpha|<1$: decays. $|\alpha|>1$: grows. $\alpha<0$: the sign flips every step.
DT frequencies repeat every 2π: $e^{j(\hat{\omega}+2\pi)n}=e^{j\hat{\omega}n}e^{j2\pi n}=e^{j\hat{\omega}n}$, because n is an integer and $e^{j2\pi n}$ is a whole number of turns.

CT $e^{j\omega t}$	DT $e^{j\hat{\omega}n}$
each ω gives a different signal	$\hat{\omega}$ and $\hat{\omega}+2\pi$ give the same signal
always periodic	periodic only if $\frac{\hat{\omega}}{2\pi}$ is rational
larger $\omega \Rightarrow$ faster	fastest at $\hat{\omega}=\pi$ $(+1,-1,+1,\ldots)$

Worked: real part of $x(t) = 2e^{j\pi/3}e^{(-1+j4)t}$

1. SPLIT THE EXPONENT
$e^{(-1+j4)t} = e^{-t}\cdot e^{j4t}$ *(shrinking part times spinning part)*

↓

2. COMBINE THE j PARTS
$e^{j\pi/3}\cdot e^{j4t} = e^{j(4t+\pi/3)}$ *(add the exponents)*

↓

3. REWRITE
$x(t) = 2e^{-t}e^{j(4t+\pi/3)}$

↓

4. EULER
$x(t) = 2e^{-t}\left[\cos(4t+\frac{\pi}{3}) + j\sin(4t+\frac{\pi}{3})\right]$

↓

5. KEEP THE REAL PART
$\mathrm{Re}\{x\} = 2e^{-t}\cos\left(4t+\frac{\pi}{3}\right)$: a damped sinusoid, envelope $2e^{-t}$

Worked: (reverse) write $5e^{-2t}\cos\left(3t-\frac{\pi}{4}\right)$ ***as*** $\mathrm{Re}\{Ce^{at}\}$

1. MATCH $|C|e^{\sigma t}\cos(\omega t+\phi)$
$|C|=5,\quad \sigma=-2,\quad \omega=3,\quad \phi=-\frac{\pi}{4}$

↓

2. BUILD C
$C = |C|e^{j\phi} = 5e^{-j\pi/4}$

↓

3. BUILD a
$a = \sigma + j\omega = -2+j3$

Worked: cosine to exponentials: $6\cos(5t)$

1. FORMULA
$\cos\theta = \dfrac{e^{j\theta}+e^{-j\theta}}{2},\quad \theta = 5t$

↓

2. MULTIPLY BY 6
$6\cdot\dfrac{e^{j5t}+e^{-j5t}}{2} = 3e^{j5t}+3e^{-j5t}$

Worked: simplify $x[n] = \cos\left(\frac{9\pi}{4}n\right)$

1. REMOVE A WHOLE TURN
$\dfrac{9\pi}{4} - 2\pi = \dfrac{9\pi}{4} - \dfrac{8\pi}{4} = \dfrac{\pi}{4}$

↓

2. SAME SAMPLES
$x[n] = \cos\left(\frac{\pi}{4}n\right)$ *($2\pi n$ is a whole number of turns)*

↓

3. PERIOD
$\dfrac{\pi/4}{2\pi} = \dfrac{1}{8} \Rightarrow N = 8$

10. the button, the switch, and the ramp

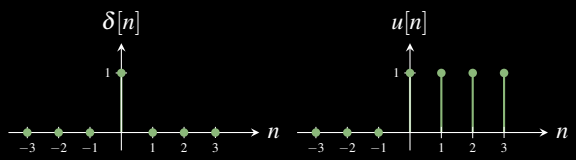
THE IDEA

- δ is a **button**: press it once at time 0, then release. On for one instant, then off.
- u is a **switch**: flip it on at time 0 and leave it on forever.
- r is a **ramp**: the running total of the switch. It climbs $0,1,2,3,\ldots$

Each one is the running total of the one before. Everything else in the course is built from these three.

DT button and DT switch:

$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n\neq 0 \end{cases} \qquad u[n] = \begin{cases} 1 & n\geq 0 \\ 0 & n<0 \end{cases}$$



Trap: The switch does not keep climbing. $u[n]$ is $0,0,1,1,1,\ldots$, not $1,2,3$. "Step" means *one* stair up, then flat. The thing that climbs is the ramp.

Energy and power of the building blocks: the button is one stem of height 1, so $E_\infty = 1^2 = 1$: an energy signal. The switch stays on forever: $E_\infty = \infty$, finite power, a power signal.

The switch is the running total of button presses

Count every press from $-\infty$ up to now:

$$u[n] = \sum_{k=-\infty}^{n} \delta[k]$$

- before $n=0$: no presses yet, so total $=0$
- at $n=0$: one press, so total $=1$
- after: no new presses, so the total stays 1

Going back the other way: a switch on at 0 *minus* a switch on at 1 is on only at 0. That is the button: $\delta[n] = u[n] - u[n-1]$.

Shifting: where is the button?

$\delta[n-k]$ puts the stem at $n=k$. To find it, ask: what n makes the inside equal 0? For $\delta[n-2]$: $n-2=0$, so $n=2$. The switch shifts the same way: $u[n-k]$ turns on at $n=k$.

Inside	Stem at	Moves	Name
$n-3$	$n=3$	right	**delay**: it happens later
$n+2$	$n=-2$	left	**advance**: it happens earlier

Trap: The sign inside is opposite to the position. And $+2$ is an *advance*, not a delay.

Building signals: switches on and off, one button per stem

Switch on, then switch off gives a **pulse**. In $u[n-a]-u[n-b]$, the first switch turns on at a; the second, subtracted, turns on at b and cancels the first.

- DT: on for $a\leq n\leq b-1$. That is $b-a$ samples.
- CT: $u(t-t_1)-u(t-t_2)$ is on for $t_1\leq t<t_2$.

Trap: The off-switch's own position is already off. $u[n]-u[n-4]$ is 0 at $n=4$: check $u[4]-u[0]=1-1=0$.

When the two switches are one step apart ($b-a=1$), the pulse shrinks to a single stem: a button.

One button per stem: a stem of height h at position k is $h\,\delta[n-k]$. Add one term per stem, joined with $+$ signs:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

At any n, every button is off except the one at $k=n$, and its weight is $x[n]$.

Worked: write $x[-1]=2$, $x[0]=-1$, $x[1]=3$, $x[3]=1$ (others 0) with buttons

1. STEM AT $n=-1$, HEIGHT 2
$2\,\delta[n-(-1)]=2\,\delta[n+1]$

↓

2. STEM AT $n=0$, HEIGHT -1
$-1\cdot\delta[n]=-\delta[n]$

↓

3. STEM AT $n=1$, HEIGHT 3
$3\,\delta[n-1]$

↓

4. STEM AT $n=3$, HEIGHT 1
$\delta[n-3]$ *($x[2]=0$, so no term for $n=2$)*

↓

5. ADD INTO ONE SIGNAL
$x[n]=2\delta[n+1]-\delta[n]+3\delta[n-1]+\delta[n-3]$

Worked: (reverse) read the stems of $x[n]=4\delta[n-2]-2\delta[n+3]$

1. FIRST TERM
$\delta[n-2]$ is 1 at $n=2$. Stem: $x[2]=4$

↓

2. SECOND TERM
$\delta[n+3]$ is 1 at $n=-3$. Stem: $x[-3]=-2$

↓

3. CHECK AT $n=2$
$4\delta[0]-2\delta[5]=4(1)-2(0)=4$ ✓

Worked: DT: $x[n]=u[n]-u[n-3]$

1. TABLE
$n=-1$: $0-0=0$. $n=0,1,2$: $1-0=1$. $n=3$: $1-1=0$. *(on from 0 through $3-1=2$)*

↓

2. AS BUTTONS
One per stem: $\delta[n]+\delta[n-1]+\delta[n-2]$

↓

3. FIRST DIFFERENCE
$u[n]-u[n-1]=\delta[n]$ and $u[n-3]-u[n-4]=\delta[n-3]$
$x[n]-x[n-1]=\delta[n]-\delta[n-3]$ *(a button at each switch)*

The CT button

In continuous time, a button pressed for "one sample" does not exist. Instead, picture a rectangle of width ε and height $\frac{1}{\varepsilon}$:

- its area is $\varepsilon\cdot\frac{1}{\varepsilon}=1$, always
- shrink $\varepsilon\to 0$: infinitely narrow, infinitely tall, area still 1

Think of a hammer tap: instant, but it delivers a definite amount of hit. That is why the CT button has area 1, while the DT button has height 1:

$$\int_{-\infty}^{\infty}\delta(t)\,dt=1$$

CT switch $u(t)$: 0 for $t<0$, 1 for $t>0$. The CT running total is an integral: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$. Going back is a derivative: $\delta(t)=\frac{du}{dt}$. A jump in a signal becomes a button in its derivative.

Trap: δ is a *function* of t, like $\sin(t)$, not a variable. $\int\delta(t)\,dt$ is *not* δ^2. You never antidifferentiate δ with a formula. You use its one fact: the area is 1.

Sifting: the camera flash

$$\int_{-\infty}^{\infty}x(t)\,\delta(t-t_0)\,dt=x(t_0)$$

Read it in English: fire a flash at t_0, and see what value of x it captures.

1. $\delta(t-t_0)$ is zero everywhere except $t=t_0$, so only $x(t_0)$ matters.
2. At that one instant, $x(t_0)$ is a constant, so it comes out of the integral.
3. What is left is $\int\delta\,dt=1$, the area.

Three questions, every time:

1. **Where does the flash sit?** Set the inside of δ to 0 and solve for t_0.
2. **Is it in the window?** Check that t_0 is inside the limits. If not, the answer is 0: the flash went off where you were not looking.
3. **What does it capture?** Plug t_0 into everything except the δ.

The final answer is a plain number, with no δ left in it.

Sampling form: $x(t)\,\delta(t-t_0)=x(t_0)\,\delta(t-t_0)$. DT: $x[n]\,\delta[n-n_0]=x[n_0]\,\delta[n-n_0]$.

Which way? — switch or button

Given → find	Do this
$u\to\delta$ (CT)	differentiate: $\delta(t)=\frac{du}{dt}$
$\delta\to u$ (CT)	integrate: $u(t)=\int_{-\infty}^{t}\delta(\tau)\,d\tau$
$u\to\delta$ (DT)	first difference: $\delta[n]=u[n]-u[n-1]$
$\delta\to u$ (DT)	running sum: $u[n]=\sum_{k=-\infty}^{n}\delta[k]$

Trap: CT uses derivatives and integrals. DT uses differences and sums.

Worked: $\int_{-2}^{6}(3t+1)\,\delta(t-4)\,dt$

1. WHERE DOES THE FLASH SIT?
$t-4=0\Rightarrow t_0=4$

↓

2. IS IT IN THE WINDOW?
Is 4 between -2 and 6? Yes ✓

↓

3. WHAT DOES IT CAPTURE?
$3(4)+1=13$

↓

4. WHY THAT IS LEGAL
At $t=4$, $3t+1$ is the constant 13: $13\int_{-2}^{6}\delta(t-4)\,dt=13\cdot 1=13$

Worked: $\int_{1}^{3}5\,\delta(t)\,dt$

1. WHERE DOES THE FLASH SIT?
The inside is t: $t_0=0$

↓

2. IS IT IN THE WINDOW?
0 is not in $[1,3]\Rightarrow$ the integral is 0 *(with limits $-\infty$ to ∞, it would be $5\cdot 1$)*

Worked: $\int_{-\infty}^{\infty}(t^2+1)\,\delta(t-2)\,dt$

1. WHERE DOES THE FLASH SIT?
$t-2=0\Rightarrow t_0=2$

↓

2. IS IT IN THE WINDOW?
2 is inside $(-\infty,\infty)$ ✓

↓

3. WHAT DOES IT CAPTURE?
$2^2+1=4+1=5$

Worked: $\int_{0}^{5}e^t\,\delta(t+1)\,dt$

1. WHERE DOES THE FLASH SIT?
$t+1=0\Rightarrow t_0=-1$

↓

2. IS IT IN THE WINDOW?
-1 is not in $[0,5]\Rightarrow$ the integral is 0

Worked: DT: $\sum_{n=-\infty}^{\infty}n^2\,\delta[n-3]$

1. WHERE IS THE BUTTON?
$n-3=0\Rightarrow n_0=3$

↓

2. WHAT DOES IT CAPTURE?
$3^2=9$

Worked: write a pulse of height 2 on $[1,4)$ with switches, then go down and back

1. SWITCH ON AT $t=1$
A jump up of 2: $+2u(t-1)$

↓

2. SWITCH OFF AT $t=4$
A jump down of 2: $-2u(t-4)$

↓

3. ADD
$x(t) = 2u(t-1) - 2u(t-4)$

↓

4. CHECK TWO TIMES
$t=2$: $2(1)-2(0)=2$ ✓ $t=5$: $2(1)-2(1)=0$ ✓

↓

5. DIFFERENTIATE (DOWN)
Each switch becomes a button: $\dfrac{dx}{dt} = 2\delta(t-1) - 2\delta(t-4)$

↓

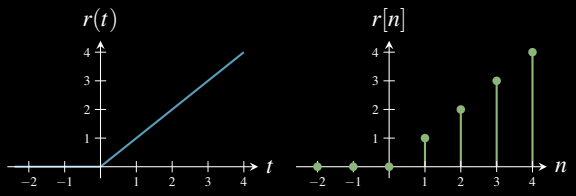
6. INTEGRATE (BACK)
Each button becomes a switch: $2u(t-1) - 2u(t-4)$ ✓

The ramp: the running total of the switch

The **unit ramp** is 0 before time 0, then climbs with slope 1:

$$r(t) = t\,u(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases} \qquad r[n] = n\,u[n] = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

It grows without limit, so it is neither an energy nor a power signal.



The chain:

$$\text{button } \delta \xrightarrow{\text{running total}} \text{switch } u \xrightarrow{\text{running total}} \text{ramp } r$$

Going back: differentiate (CT) or take the first difference (DT).

Other ways to write the ramp:

Form	Formula
CT: integral of the switch	$r(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$
CT: derivative of the ramp	$\dfrac{dr}{dt} = u(t)$
DT: running sum of the switch	$r[n] = \sum_{k=-\infty}^{n-1} u[k]$
DT: first difference	$r[n] - r[n-1] = u[n-1]$
DT: sum of shifted switches	$r[n] = \sum_{k=1}^{\infty} u[n-k]$
DT: sum of buttons	$r[n] = \sum_{k=0}^{\infty} k\,\delta[n-k]$

Trap: The DT sum stops at $n-1$, not n. $r[1] = 1$: only $u[0]$ is added. With n as the top, you get $r[n+1]$.

Shifted and scaled: $A\,r(t-t_0)$ is 0 before t_0, then climbs with slope A.

Graphs from ramps and switches: walk left to right. At each corner, add a ramp weighted by the *change in slope*. At each jump, add a switch weighted by the *size of the jump*.

Worked: check $r[n] = \sum_{k=-\infty}^{n-1} u[k]$ at $n = 0, 1, 3$

1. $n=0$
Sum of $u[k]$ for $k \leq -1$: all $0 \Rightarrow 0 = r[0]$ ✓

↓

2. $n=1$
$k \leq 0$: only $u[0] = 1 \Rightarrow 1 = r[1]$ ✓

↓

3. $n=3$
$k \leq 2$: $u[0]+u[1]+u[2] = 3 = r[3]$ ✓

Worked: first difference of the ramp

1. $n=0$
$r[0] - r[-1] = 0 - 0 = 0$

↓

2. $n=1$
$r[1] - r[0] = 1 - 0 = 1$

↓

3. $n=2$
$r[2] - r[1] = 2 - 1 = 1$

↓

4. PATTERN
0 up to $n=0$, then 1 from $n=1$ on $\Rightarrow u[n-1]$

Worked: write a rise from 0 to 2 on $[0,2]$, then flat at 2, with ramps

1. CORNER AT $t=0$
Slope changes $0 \to 1$: change $+1 \Rightarrow +r(t)$

↓

2. CORNER AT $t=2$
Slope changes $1 \to 0$: change $-1 \Rightarrow -r(t-2)$

↓

3. ADD
$x(t) = r(t) - r(t-2)$

↓

4. CHECK $t=1$
$r(1) - r(-1) = 1 - 0 = 1$ ✓

↓

5. CHECK $t=5$
$r(5) - r(3) = 5 - 3 = 2$ ✓ *(flat at 2)*

Worked: a triangle: 0 at $t=0$, peak 1 at $t=1$, back to 0 at $t=2$

1. CORNER AT $t=0$
Slope $0 \to 1$: change $+1 \Rightarrow +r(t)$

↓

2. CORNER AT $t=1$
Slope $1 \to -1$: change $-2 \Rightarrow -2r(t-1)$

↓

3. CORNER AT $t=2$
Slope $-1 \to 0$: change $+1 \Rightarrow +r(t-2)$

↓

4. ADD
$x(t) = r(t) - 2r(t-1) + r(t-2)$

↓

5. CHECK $t=3$
$3 - 2(2) + 1 = 0$ ✓ *(the slopes add to $1-2+1=0$)*

Worked: (reverse) sketch $x(t) = 2r(t-1) - 2r(t-3)$

1. BEFORE $t = 1$
Both ramps are $0 \Rightarrow x = 0$

↓

2. FROM $t = 1$ TO $t = 3$
Only the first ramp: slope 2. At $t = 3$: $2(3-1) = 4$

↓

3. AFTER $t = 3$
Slope $2 - 2 = 0$: flat at 4

↓

4. CHECK $t = 5$
$2(4) - 2(2) = 8 - 4 = 4$ ✓

Worked: a switch plus a ramp: a jump of 3 at $t = 0$, then slope -1 until it reaches 0

1. JUMP AT $t = 0$
$+3u(t)$

↓

2. CORNER AT $t = 0$
Slope $0 \to -1 \Rightarrow -r(t)$

↓

3. CORNER WHERE IT REACHES 0
$3 - t = 0$ at $t = 3$. Slope $-1 \to 0 \Rightarrow +r(t-3)$

↓

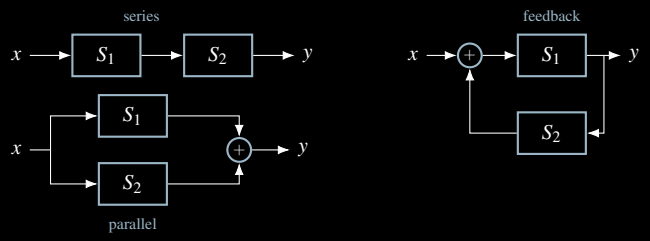
4. ADD
$x(t) = 3u(t) - r(t) + r(t-3)$

11. systems and interconnections

THE IDEA
A **system** is a machine: a signal goes in, a different signal comes out, $x \to y$. A speaker, a filter, a bank account. A **difference equation** is the machine's recipe: how to cook up $y[n]$ from inputs, and sometimes from its own past outputs.

Machines can be hooked together three ways:

- **Series** (cascade): an assembly line. The output of S_1 is the input of S_2.
- **Parallel**: x goes into both machines at once. Their outputs add.
- **Feedback**: the output loops back through S_2 and adds into the input. The machine hears its own past.



Worked: series: S_1: $w[n] = 2x[n]$, then S_2: $y[n] = w[n] - w[n-1]$

1. OUTPUT OF S_1
$w[n] = 2x[n]$

↓

2. SHIFT IT FOR S_2
$w[n-1] = 2x[n-1]$ *(replace every n with $n-1$)*

↓

3. SUBSTITUTE INTO S_2
$y[n] = 2x[n] - 2x[n-1]$

Worked: parallel: S_1: $2x[n]$ and S_2: $x[n-1]$

1. EACH OUTPUT
$y_1[n] = 2x[n], \quad y_2[n] = x[n-1]$

↓

2. ADD
$y[n] = 2x[n] + x[n-1]$

Worked: feedback: $y[n] = x[n] + 0.5\,y[n-1]$, input: press the button, $x[n] = \delta[n]$, $y[-1] = 0$

1. $n = 0$
$y[0] = \delta[0] + 0.5\,y[-1] = 1 + 0.5(0) = 1$ *(the press)*

↓

2. $n = 1$
$y[1] = \delta[1] + 0.5\,y[0] = 0 + 0.5(1) = 0.5$ *(button released; only the echo)*

↓

3. $n = 2$
$y[2] = 0 + 0.5(0.5) = 0.25$

↓

4. $n = 3$
$y[3] = 0 + 0.5(0.25) = 0.125$

↓

5. PATTERN
$y[n] = (0.5)^n\,u[n]$ *(each echo is half the last)*

DT systems as difference equations

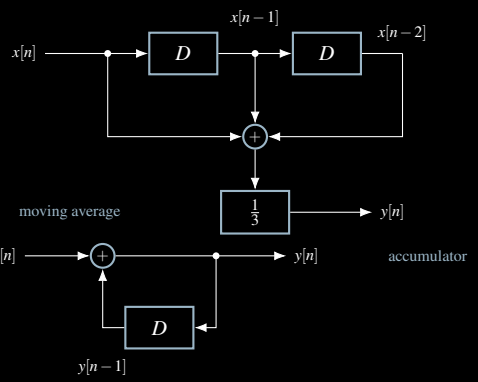
System	Difference equation	What it does
unit delay	$y[n] = x[n-1]$	repeats the input one step late
gain	$y[n] = a\,x[n]$	volume knob
first difference	$y[n] = x[n] - x[n-1]$	change since the last sample
3-point moving average	$y[n] = \dfrac{x[n] + x[n-1] + x[n-2]}{3}$	smooths
accumulator	$y[n] = \sum_{k=-\infty}^{n} x[k]$	running total
accumulator, recursive	$y[n] = y[n-1] + x[n]$	same running total
bank balance	$y[n] = 1.01\,y[n-1] + x[n]$	1% interest plus deposit

The accumulator is the "running total" machine. Feed it the button and out comes the switch. Feed it the switch and out comes a ramp.

Nonrecursive (FIR): $y[n]$ uses inputs only. **Recursive** (IIR): $y[n]$ also uses its own past outputs $y[n-1], \ldots$, and needs a starting value such as $y[-1] = 0$.

Block diagram parts:

Part	What it does
adder (circle, $+$)	output = sum of its inputs
gain (box with a)	output = a·input
unit delay (box D)	a one-step memory: in $x[n]$, out $x[n-1]$
branch point (dot)	copies a signal to two places



Which way? — equation or block diagram

Given → find	Do this
diagram → equation	name the signal on each wire; write the adder's output as the sum of its inputs
equation → diagram	one D per shift ($n-1$, $n-2$, …); one gain per coefficient; one adder for the sum
equation → output values	put in $n = 0, 1, 2, \ldots$ in order, one step at a time

Trap: A delay on the output side means feedback. If a D takes $y[n]$ back into the adder, the equation has $y[n-1]$: it is recursive.

Worked: (diagram to equation) read the moving-average diagram

1. NAME THE WIRES
After the first D: $x[n-1]$. After the second D: $x[n-2]$.

↓

2. ADDER OUTPUT
$x[n]+x[n-1]+x[n-2]$

↓

3. GAIN
$y[n]=\frac{1}{3}(x[n]+x[n-1]+x[n-2])$

Worked: (equation to diagram) $y[n]=2x[n]-0.5\,y[n-1]$

1. COUNT THE SHIFTS
One $y[n-1]\Rightarrow$ one D, on the output side (feedback)

↓

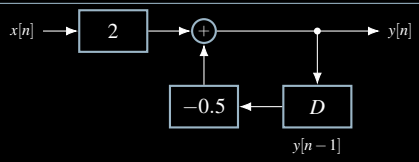
2. COUNT THE COEFFICIENTS
Gain 2 on the input. Gain -0.5 after the D.

↓

3. ONE ADDER
Its inputs: $2x[n]$ and $-0.5\,y[n-1]$. Its output: $y[n]$.

↓

4. DRAW
$x\to[2]\to(+)\to y$. From y: a dot $\to[D]\to[-0.5]\to$ back into $(+)$.



Worked: run the moving average with $x[0]=3$, $x[1]=6$, $x[2]=9$, ***others*** 0

1. $n=0$
$y[0]=\frac{1}{3}(x[0]+x[-1]+x[-2])=\frac{1}{3}(3+0+0)=1$

↓

2. $n=1$
$y[1]=\frac{1}{3}(6+3+0)=\frac{9}{3}=3$

↓

3. $n=2$
$y[2]=\frac{1}{3}(9+6+3)=\frac{18}{3}=6$

↓

4. $n=3$
$y[3]=\frac{1}{3}(0+9+6)=\frac{15}{3}=5$

↓

5. $n=4$
$y[4]=\frac{1}{3}(0+0+9)=3$

↓

6. $n=5$ **AND LATER**
All three inputs are $0\Rightarrow y=0$

Worked: run the accumulator $y[n]=y[n-1]+x[n]$ ***with the switch*** $x[n]=u[n]$, $y[-1]=0$

1. $n=0$
$y[0]=y[-1]+u[0]=0+1=1$

↓

2. $n=1$
$y[1]=y[0]+u[1]=1+1=2$

↓

3. $n=2$
$y[2]=2+1=3$

↓

4. PATTERN
$y[n]=(n+1)\,u[n]=r[n+1]$ *(the running total of the switch is a ramp)*

Worked: (both forms agree) accumulator as a sum, $x[n]=u[n]$, ***at*** $n=2$

1. SUM FORM
$y[2]=\sum_{k=-\infty}^{2}u[k]=u[0]+u[1]+u[2]$

↓

2. ADD
$1+1+1=3$, the same as the recursive form ✓

12. system properties

THE IDEA
Properties are yes/no questions you ask about a machine. Each one is separate: a machine can pass one and fail another.

Property	In plain English
memoryless	It only looks at *now*. No looking back, no looking ahead.
causal	It cannot see the future. It may remember the past.
stable (BIBO)	It does not blow up. If the input stays under a ceiling, so does the output.
invertible	You can undo it. Different inputs give different outputs, so you can solve for x.
time invariant	It ignores the clock. Delay the input, and the output is the same, just delayed.
linear	Scale in $\to$ scale out, and add in $\to$ add out. If $x_1\to y_1$ and $x_2\to y_2$, then $ax_1+bx_2\to ay_1+by_2$.

Quick red flags

Trap: Linearity: put in $x=0$. A linear machine must give $y=0$. If it gives anything else (like the $+3$ in $2x+3$), it is not linear.
Trap: Time invariance: an n or t *outside* of $x(\cdot)$, like $n\,x[n]$, means the machine reads the clock. Scaling or flipping inside, like $x(2t)$ or $x(-t)$, also fails.
Trap: Causality: look for an *advance*, like $x[n+1]$. And $y(t)=x(-t)$ is not causal: at $t=-1$ it needs $x(1)$, a later time.
Trap: Stability: an n multiplying x, or a sum with no end, can blow up.

Worked: memory and causality of $y[n]=x[n]+x[n-1]$ ***and*** $y[n]=x[n+1]$

1. $y[n]=x[n]+x[n-1]$
It uses $x[n-1]$, a past value. has memory, causal

↓

2. $y[n]=x[n+1]$
At $n=0$ it needs $x[1]$, a future value. not causal

Worked: stability of $y[n]=\sum_{k=-\infty}^{n}x[k]$

1. PICK A CAPPED INPUT
The switch, $x[n]=u[n]$. It never goes above 1.

↓

2. COMPUTE THE OUTPUT
$y[n]=\sum_{k=0}^{n}1=n+1$ for $n\geq 0$ *(the running total of the switch: a ramp)*

↓

3. IS IT CAPPED?
$n+1$ grows without limit $\Rightarrow$ not stable

Worked: invertibility of $y=2x$ ***and*** $y=x^2$

1. $y=2x$
Solve: $x=\frac{y}{2}\Rightarrow$ invertible

↓

2. $y=x^2$
$x=1$ and $x=-1$ both give $y=1\Rightarrow$ not invertible *(squaring throws away the sign)*

Time invariance: two paths, then compare

Path 1: delay the input first, then run the machine. **Path 2**: run the machine, then delay the output. If both give the same thing, the machine ignores the clock.

Worked: time invariance of $y[n] = nx[n]$

1. PATH 1: DELAY IN, THEN MACHINE
New input $x_2[n] = x[n-n_0]$. Output: $y_2[n] = nx[n-n_0]$

↓

2. PATH 2: MACHINE, THEN DELAY OUT
Replace each n in $y[n] = nx[n]$ with $n-n_0$: $y[n-n_0] = (n-n_0)x[n-n_0]$

↓

3. COMPARE
$nx[n-n_0] \neq (n-n_0)x[n-n_0] \Rightarrow$ time varying *(the lone n reads the clock)*

Worked: time invariance of $y(t) = x(2t)$

1. PATH 1: DELAY IN, THEN MACHINE
$x_2(t) = x(t-t_0)$. $y_2(t) = x_2(2t) = x(2t-t_0)$ *(put $2t$ where t was in x_2)*

↓

2. PATH 2: MACHINE, THEN DELAY OUT
$y(t-t_0) = x\big(2(t-t_0)\big) = x(2t-2t_0)$

↓

3. COMPARE
$x(2t-t_0) \neq x(2t-2t_0) \Rightarrow$ time varying

Worked: time invariance of $y[n] = x[n-1]$

1. PATH 1: DELAY IN, THEN MACHINE
$y_2[n] = x_2[n-1] = x[n-1-n_0]$

↓

2. PATH 2: MACHINE, THEN DELAY OUT
$y[n-n_0] = x[n-n_0-1]$

↓

3. COMPARE
The same $\Rightarrow$ time invariant

Linearity: combine first, or combine after?

Path 1: mix the inputs ($ax_1 + bx_2$), then run the machine. **Path 2**: run each input separately, then mix the outputs ($ay_1 + by_2$). Linear means both paths agree.

Worked: linearity of $y[n] = 2x[n] + 3$

1. QUICK TEST
$x = 0$ gives $y = 3 \neq 0 \Rightarrow$ not linear. The full test confirms it:

↓

2. PATH 1: MIX, THEN MACHINE
$x_3 = ax_1 + bx_2$. $y_3 = 2(ax_1 + bx_2) + 3 = 2ax_1 + 2bx_2 + 3$

↓

3. PATH 2: MACHINE, THEN MIX
$ay_1 + by_2 = a(2x_1 + 3) + b(2x_2 + 3) = 2ax_1 + 2bx_2 + 3a + 3b$

↓

4. COMPARE
$3 \neq 3a + 3b$ for most $a, b \Rightarrow$ not linear *(linear plus a constant: incrementally linear)*

Worked: linearity of $y[n] = nx[n]$

1. PATH 1: MIX, THEN MACHINE
$y_3 = n\big(ax_1[n] + bx_2[n]\big) = anx_1[n] + bnx_2[n]$

↓

2. PATH 2: MACHINE, THEN MIX
$ay_1 + by_2 = anx_1[n] + bnx_2[n]$

↓

3. COMPARE
The same $\Rightarrow$ linear *(linear, but time varying: the properties are separate)*

Stability with a ceiling B — the procedure

THE IDEA
BIBO means **bounded in, bounded out**. "Bounded" just means there is a ceiling: some finite number B the input never goes above in size, $|x[n]| \leq B < \infty$. Stable means the output has a ceiling too. This is Lesson 1 again: it asks whether a signal's size stays finite.

To show stable:

1. Assume the input is capped: $|x[n]| \leq B$ for every n. Shifted inputs are capped too: $|x[n-1]| \leq B$.
2. Take the absolute value of the equation.
3. Split it with $|a+b| \leq |a| + |b|$ and $|cx| = |c|\,|x|$.
4. Replace each $|x[\cdot]|$ with B.
5. If you are left with a finite number (no n left, no endless sum), the output is capped: stable.

To show unstable: find one capped input (for example $x[n] = 1$, or the switch $u[n]$) whose output grows without limit. One counterexample is enough.

Trap: An n multiplying x breaks the ceiling: $|nx[n]| \leq |n|B$ grows with n.

Trap: An endless sum can break it: the accumulator adds another B at every step.

Worked: is $y[n] = 2x[n] + x[n-1]$ stable?

1. ASSUME A CEILING
$|x[n]| \leq B$ for every n, so $|x[n-1]| \leq B$ too

↓

2. ABSOLUTE VALUE
$|y[n]| = |2x[n] + x[n-1]|$

↓

3. SPLIT THE SUM
$\leq 2|x[n]| + |x[n-1]|$

↓

4. PUT IN B
$\leq 2B + B = 3B$

↓

5. CONCLUDE
$3B$ is finite $\Rightarrow$ stable

Worked: is the 3-point moving average stable?

1. ABSOLUTE VALUE AND SPLIT
$|y[n]| \leq \frac{1}{3}\big(|x[n]| + |x[n-1]| + |x[n-2]|\big)$

↓

2. PUT IN B
$\leq \frac{1}{3}(B + B + B) = B \Rightarrow$ stable

Worked: is $y[n] = nx[n]$ stable?

1. TRY A CAPPED INPUT
$x[n] = 1$ for all n. $B = 1$

↓

2. OUTPUT
$y[n] = n \cdot 1 = n$

↓

3. CONCLUDE
n grows without limit $\Rightarrow$ not stable *(one counterexample is enough)*

Worked: is $y[n] = e^{x[n]}$ stable?

1. ASSUME A CEILING
$-B \leq x[n] \leq B$

↓

2. $e^{(\cdot)}$ INCREASES
$e^{-B} \leq e^{x[n]} \leq e^{B}$

↓

3. CONCLUDE
$|y| \leq e^{B}$, finite $\Rightarrow$ stable

Worked: (recursive) is $y[n] = 0.5\,y[n-1] + x[n]$ stable, with $y[-1] = 0$?

1. UNROLL THE RECURSION

$y[0] = x[0]$. $y[1] = x[1] + 0.5x[0]$. $y[2] = x[2] + 0.5x[1] + 0.25x[0]$. *(each past input echoes, halved each step)*

↓

2. PATTERN

$y[n] = \sum_{k=0}^{n} (0.5)^k\, x[n-k]$

↓

3. ABSOLUTE VALUE AND B

$|y[n]| \le \sum_{k=0}^{n} (0.5)^k B \le B \sum_{k=0}^{\infty} (0.5)^k$

↓

4. GEOMETRIC SUM

$$\sum_{k=0}^{\infty} (0.5)^k = \frac{1}{1-0.5} = 2$$

↓

5. CONCLUDE

$|y| \le 2B \Rightarrow$ stable *(with 1.01 in place of 0.5, the echoes grow and the sum has no limit: unstable)*

Properties of common DT systems

System	Mless	Causal	Linear	TI	Stable
$x[n-1]$	no	yes	yes	yes	yes
$x[n] - x[n-1]$	no	yes	yes	yes	yes
moving average	no	yes	yes	yes	yes
accumulator	no	yes	yes	yes	no
$n\,x[n]$	yes	yes	yes	no	no
$x[n]^2$	yes	yes	no	yes	yes
$2x[n] + 3$	yes	yes	no	yes	yes
$x[n+1]$	no	no	yes	yes	yes
$x[2n]$	no	no	yes	no	yes
$x[-n]$	no	no	yes	no	yes
$0.5\,y[n-1] + x[n]$	no	yes	yes	yes	yes

Mless = memoryless. TI = time invariant. The recursive system starts at rest ($y[-1] = 0$).

Trap: The properties are separate. A machine can be linear but not time invariant ($n\,x[n]$), or stable but not causal ($x[n+1]$).